

StatRotationLoadedTime [SimTalk] - Turntable

Returns the total time during which the *Turntable* designated by <Path> was rotating with a *MU* being located on the table.

Type

Read-only attribute

Syntax

```
<Path>.StatRotationLoadedTime → time
```

Return Value

The return value has the data type *time*.

Example

```
print MyTurntable.StatRotationLoadedTime
```

See also

Statistics report, [Rotation Time](#)

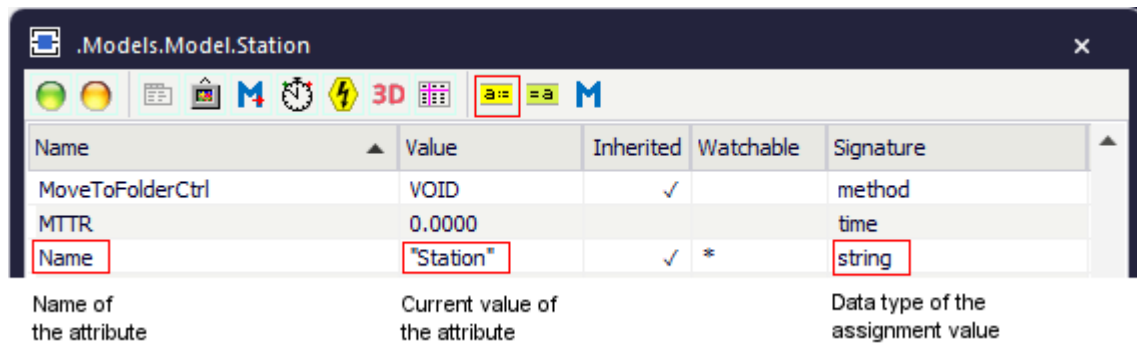
Attributes of the Turntable

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The *Turntable* provides:

- The *attributes* listed in the table of contents to the left.
- The [Attributes of All Objects](#).
- The [Attributes of the Material Flow Objects](#).

To view all of the *methods*, *read-only attributes*, and *attributes* of the object, open the window [Show Attributes and Methods](#). The figure below illustrates the information using the example of the object *Station*.



- Select **Show Attributes and Methods** on the context menu of the **Class Library** to show the *methods*, *read-only attributes*, and *attributes* of the selected [Class \[general description\]](#).
- Press the **F8** key or click **Show Attributes and Methods** on the **Home** ribbon tab of the *Frame* into which you inserted an instance to show the *methods*, *read-only attributes*, and *attributes* of the selected [Instance \[general description\]](#).

You can set the value of an attribute and you can get its value, either with the check boxes, the text boxes and drop-down lists in the dialog windows or by assigning values to the respective attributes.

- To **set the value** of an attribute, you might, for example, type:

```
MyTurntable.GoToDefaultPosition := false
```

- To **get the value** of an attribute, you might, for example, type:

```
print MyTurntable.GoToDefaultPosition
posit := MyStation.Cont.XPos
```

AutomaticStop [SimTalk] - Turntable

Automatically stops the *Turntable* designated by<Path>. This means that it sets the current speed to 0 if it does not transport a *MU*. This might, for example, be the case if it is empty or if it is blocked because a *MU* cannot leave it.

Remarks

If the speed of the *Turntable* is 0, the **Energy State** changes to **operational**.

Type

Attribute

Syntax

```
<Path>.AutomaticStop:boolean
```

Assignment Value

You can assign a value of data type *boolean*.

Example

```
MyTurntable.AutomaticStop := true
```

See also

[Automatic Stop \[check box\] - Turntable](#)

[Operational \[energy\]](#)

DefaultAngle [SimTalk] - Turntable

Sets the rotation angle in degrees to which the side of the end point of the *Turntable* designated by <Path> rotates.

Remarks

DefaultAngle applies when you set **Go to Default Position**/*GoToDefaultPosition* to `true`.

To rotate the *Turntable* to the side on which the start point is located, add 180° to this angle.

Type

Attribute

Syntax

```
<Path>.DefaultAngle:real
```

Assignment Value

You can assign a value of data type *real*.

Example

```
MyTurntable.DefaultAngle := 5
```

SimTalk

[GoToDefaultPosition \[SimTalk\] - PickAndPlace](#)

See also

[Default Angle \[text box\] - Turntable](#)

[Go to Default Position \[check box\] - Turntable](#)

GoToDefaultPosition [SimTalk] - Turntable

Returns the *Turntable* designated by <Path> to its default position (`true`) or not (`false`).

Remarks

GoToDefaultPosition applies when the part has left the *Turntable* and if no new part is ready to be rotated.

Note

If you set the attribute to `true`, the *Turntable* uses the attribute *DefaultAngle*, when you reset the model.

The attribute *DefaultAngle* sets the actual angle.

Type

Attribute

Syntax

```
<Path>.GoToDefaultPosition:boolean
```

Assignment Value

You can assign a value of data type *boolean*.

Example

```
MyTurntable.GoToDefaultPosition := true
```

SimTalk

[DefaultAngle \[SimTalk\] - Turntable](#)

See also

[Go to Default Position \[check box\] - Turntable](#)

[Default Angle \[text box\] - Turntable](#)

Length [SimTalk] - Turntable

Sets the **Length** of the *Turntable* designated by <Path>.

Type

Attribute

Syntax

```
<Path>.Length: length
```

Watchable

The attribute is **watchable**.

Assignment Value

You can assign a value of data type *length*.

Note

In *SimTalk* 2.0 you can specify the length units m, mm, km, cm, yd, ft, and in.

Type in the unit directly after the values, without a separating blank space, for example 10m or 10.2m.

You can specify the unit for *floating point values* and for *integer values*.

Example

```
MyTurntable.Length := 10.0m
```

See also

[Length \[text box\] - Turntable](#)

MURotationAttribute [SimTalk]

Sets name of the user-defined attribute of the *MU* which triggers its rotation on the *Turntable* designated by <Path>.

Remarks

MURotationAttribute can be an attribute of data type *boolean* or a *Method*, which returns a value of data type *boolean*. *MURotationAttribute* will only be evaluated, when you select **Any** as the **Side** in the **Exit Angle Table**.

Type

Attribute

Syntax

```
<Path>.MURotationAttribute:string
```

Assignment Value

You can assign a value of data type *string*.

Example

```
MyTurntable.MURotationAttribute := "MyRotationAttribute"
```

See also

[Exit Angle Table](#)

[MU Leaves Backwards Depends On](#)

RotateWhen [SimTalk]

Sets, when the *Turntable* designated by <Path> rotates toward its target station.

Type

Attribute

Syntax

```
<Path>.RotateWhen:string
```

Assignment Value

You can assign a value of data type *string*.

You can specify:

- "Completely entered", i.e., if the *MU* has completely entered the *Turntable*.
- "Rotation point reached", i.e., if the *MU* has reached the center of rotation on the table.
- "Centered", i.e., if both ends of the *MU* have the same distance from the ends of the turntable.
- "User-defined with Sensor". Here the *Turntable* does not rotate automatically, but only when the *MU* has reached a user-defined sensor. In the **Sensor Control**, you have to tell the *Turntable* to call the method [setDestination](#)

Example

```
MyTurntable.RotateWhen := "completely entered"
```

SimTalk

[setDestination \[SimTalk\] - Turntable](#)

See also

[Strategy \[drop-down list\] - Turnplate](#)

[Rotation Point \[text box\]](#)

RotationPoint [SimTalk]

Sets the position of the point around which the *Turntable* designated by <Path> rotates.

Type

Attribute

Syntax

```
<Path>.RotationPoint:length
```

Assignment Value

You can assign a value of data type *length*.

Specify a value between 0 and the length of the *Turntable*. 0 designates the point at which you start inserting the *Turntable*.

Example

```
MyTurntable.RotationPoint := 4
```

See also

[Rotation Point \[text box\]](#)

RotationTimePer90Degrees [SimTalk] - Turntable

Sets the time it takes the *Turntable* designated by <Path> to rotate by 90 degrees.

Type

Attribute

Syntax

```
<Path>.RotationTimePer90Degrees:time
```

Assignment Value

You can assign a value of data type *time*.

Example

```
MyTurntable.RotationTimePer90Degrees := 2
```

See also

[Rotation Time per 90° \[text box\] - Turntable](#)

Speed [SimTalk] - Turntable

Sets the speed, with which the *Turntable* designated by <Path> rotates.

Type

Attribute

Syntax

```
<Path>.Speed : speed
```

Watchable

The attribute is [watchable](#).

Assignment Value

You can assign a value of data type *speed*.

Specify -1 for an infinite speed.

Note

In *SimTalk* 2.0 you can specify the speed units mps, fps, kmh, and mph.

Type in the unit directly after the number, without a separating blank space, for example 100kmh or 100.5kmh.

```
var len := 1.0ft  
var s : speed := 10.5m / 1:30  
var x : length := 3m
```

Example

```
MyTurntable.Speed := 1.5
```

See also

[Conveyor Speed \[text box\]](#)

TargetCtrl [SimTalk] - Turntable

Designates a *Method* object of the object designated by <Path>.

Remarks

Plant Simulation calls the *Method* as soon as the *MU* has completely moved onto the *Turntable* or has reached the center of rotation.

In the **Target Control** you determine the successor to which the *Turntable* designated by <Path> moves the *MU*.

As opposed to the **Exit Control**, the *MU* is not ready to exit the object yet. You have to set the target object in the method *setDestination*.

Type

Attribute

Syntax

```
<Path>.TargetCtrl:method
```

Assignment Value

You can assign a value of data type *method*.

Examples

```
if @.Name = "PartA"
// the identifier "@" points to the MU, the part in our case,
// which triggers the call of the current method in a control
    ?.setDestination(ConveyorPartA)
    // the identifier "?" points to the object which
    // calls the method, the turntable in our case
else
    ?.setDestination(ConveyorPartB)
end
```

```
MyTurntable.TargetCtrl := &myTargetControl
```

SimTalk

[setDestination \[SimTalk\] - Turntable](#)

See also

[Target Control \[Turntable\]](#)

Width [SimTalk] - Turntable

Sets the width of the *Turntable* designated by <Path>.

Type

Attribute

Syntax

```
<Path>.Width:length
```

Watchable

The attribute is [watchable](#).

Assignment Value

You can assign a value of data type *length*.

Note

In *SimTalk* 2.0 you can specify the length units m, mm, km, cm, yd, ft, and in.

Type in the unit directly after the values, without a separating blank space, for example 10m or 10.2m.

You can specify the unit for *floating point values* and for *integer values*.

Example

```
MyTurntable.Width := 2 // meters
```

See also

[Width \[text box\] - Turntable](#)

Turnplate

Turnplate [object]

Use the object *Turnplate* for modeling a rotating platform, which rotates the loaded parts and ensures the uniform orientation of the leaving parts.

Image



Description

A typical example is in the package shipping industry where all packages have to be rotated to a uniform direction so that a scanner can automatically read the bar code holding the address information. The *Turnplate* has a capacity of one, i.e., only one *MU* can be located on it at any one time. The *MU* moves onto the *Turnplate* and the *Turnplate* starts rotating when the booking point of the *MU* has reached the center of rotation of the *Turnplate*. When the rotation is finished, the *MU* exits the *Turnplate*. The conveying direction on the *Turnplate* is unidirectional, i.e., the *MU* cannot be conveyed forward and then backward.

The length of the *MU* must not be longer than the **Length** of the *Turnplate* itself.

The center of rotation is by default located in the center of the *Turnplate*. The rotation takes up a certain amount of time. You can type in the rotation time per rotation steps of 90 degrees. If you type in a rotation time of 0, the *MU* is rotated instantaneous without using up any time at all.

The rotation angle that you type in should be a multiple of 90 degrees. If you type in an angle other than that, *Plant Simulation* rounds this angle to the next angle that is divisible by 90. You can also type in a value greater than 360 degrees as long as it is divisible by 90. This way the *Turnplate* can rotate the *MU* several times to simulate packing machines, such as shrink wrappers, etc. By default, the *Turnplate* rotates 90 degrees clockwise.

After the turnplate has rotated the *MU* and it has left it, the *Turnplate* returns to its starting position.

You can select different configurations for the *Turnplate* on the tab **Appearance of the Length-oriented Objects**.